

Sai Sumanth Badugu

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Summary

Data Engineer with 4+ years of experience building scalable **ETL pipelines**, real-time streaming systems, and cloud-based data architectures using **AWS**, **GCP**, and **Spark** ecosystems. Proven success integrating multi-source data with **AWS Glue**, **PySpark**, and **Kafka**, reducing processing latency by up to **56%** and driving higher accuracy and efficiency across enterprise data platforms. Skilled in transforming complex datasets into actionable insights through **Power BI**, **Tableau**, empowering organizations with data-driven decision-making.

Education

Master of Science in Analytics, Saint Louis University, St. Louis, MO — **CGPA: 3.85/4**

Aug 2023 – May 2025

Coursework: Analytics, Data Science, Cloud Security, Visualization, Project Management, Machine Learning, Big Data Programming

Technical Skills

- **Programming & Data:** Python, SQL, PL/SQL, PySpark, Pandas, NumPy
- **Visualization & Databases:** Power BI, Tableau, Looker, Excel, PostgreSQL, MySQL
- **Cloud & Tools:** Snowflake, Databricks, Airflow, AWS (EC2, S3, Glue), Azure (ADF, Synapse, Data Lake, DevOps), Docker
- **Streaming & Pipelines:** Kafka, Delta Lake, ETL Development, Real-Time Monitoring
- **ETL & Development Tools:** Informatica, SSIS, Datastage, Apache NiFi, MATLAB, Terraform, REST API, CI/CD, Git, JIRA
- **Certifications:** Microsoft Fabric Analytics Engineer, Azure Data Scientist Associate, Power BI Data Analyst Associate

Experience

Data Engineer – Mspring Info Tech, India

Jul 2021 – Jun 2023

- **Cross-Database Integration:** Designed and implemented end-to-end, real-time, optimized, and scalable data ingestion from **Oracle DB**, **Cosmos DB**, and **MS SQL Server** using **AWS Glue**, **Redshift**, and **Lambda**, boosting processing efficiency by **56%** and cutting storage costs by **16%** through automated workflows.
- **ETL Development:** Built high-performance ETL pipelines using **Informatica**, **SSIS**, **Datastage**, and **Apache NiFi**, integrating **Hadoop** and **Spark** within **Snowflake** and **Redshift** environments. Increased data throughput by **49%** and reduced batch delays, ensuring faster and more reliable analytics.
- **Scalable Data Processing & Security:** Engineered distributed pipelines with **AWS Glue**, **PySpark**, **Scala**, and **Kinesis**, improving execution speed by **22%**. Introduced **Amazon S3 encryption**, reducing data risk by **65%** and strengthening compliance with security standards.
- **Cloud Infrastructure & Automation:** Automated **AWS EC2**, **RDS**, and **IAM** provisioning with **Terraform**, reducing setup time by **40%** and ensuring scalable, repeatable infrastructure deployments that enhanced reliability.
- **Data Visualization & BI:** Built interactive dashboards with **Tableau**, **Power BI**, and **Amazon QuickSight**, enabling clearer insights for stakeholders and improving engagement by **44%**.

Junior Data Analyst – Huddl.ai, India

Sep 2019 – May 2021

- **ETL Design & Big Data Integration:** Developed and tuned ETL pipelines using **Apache NiFi**, **Sqoop**, **Spark**, **Kafka**, and **Ray**, boosting data throughput by **49%**. This project deepened my appreciation for well-structured data architecture and its impact on scalability.
- **Pipeline Automation & Machine Learning:** Automated workflows with **Airflow** and **Oozie**, integrating **TensorFlow** and **scikit-learn** within **Spark** pipelines to reduce processing time by **35%**. Implemented **OpenLineage** and **Great Expectations** to reinforce transparency and data reliability.
- **Scalable Processing & Big Data Technologies:** Created distributed data solutions using **AWS Glue**, **PySpark**, **Scala**, and **Kinesis**. Leveraged **AWS Data Pipelines** and **Elasticsearch** to improve performance by **22%** while preserving data integrity in **S3**.
- **Data Warehousing & User Interfaces:** Enhanced query efficiency using **Databricks** while developing intuitive web interfaces that improved data accessibility by **30%** and boosted user engagement by **72%**.
- **Data Visualization:** Created dashboards in **Tableau**, **Amazon QuickSight**, and **Cognos**, reducing decision-making time by **66%**. Utilized **Polars** and **DQDL** for robust data validation and analytical consistency.

Projects

Healthcare Data Integration and Analytics

- Engineered robust ETL pipelines and relational data models on **Google Cloud Platform (GCP)** to integrate and normalize **100K+ synthetic Electronic Health Records (EHR)** across patients, encounters, and claims with **HIPAA-compliant** accuracy. Delivered interactive **Power BI** dashboards that uncovered chronic disease patterns, provider workload distribution, and insurance coverage trends, driving data-informed decision-making for healthcare operations.

Real-Time AI Generated User Data Streaming with Kafka and Airflow

- Built a real-time streaming engine with **Kafka**, **Airflow**, and **PySpark** to process continuous AI-generated user data into **Amazon S3**, enabling SQL-based behavior analytics via **Athena**.